

SOP-ELEC-500: Electrical Safe Work Practices and Arc Flash Protection

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Section 1. Purpose and Scope

Establishes safe work practices for work on or near energized electrical equipment above 50 volts, including arc flash risk assessment and PPE selection.

Section 2. Energized Work Permit

Clause 2.1 — Work on energized electrical equipment is prohibited unless de-energization is infeasible (e.g., would create a greater hazard, or is technically impossible for diagnostic testing) and an Energized Electrical Work Permit is approved by the Electrical Safety Department.

Clause 2.2 — Every energized work permit requires a documented arc flash risk assessment specific to the equipment and task, not a generic facility-wide value.

Section 3. Arc Flash PPE Categories

Clause 3.1 — PPE category is determined from the equipment's arc flash label (incident energy in cal/cm²) per the facility's arc flash study, not estimated in the field.

Clause 3.4 — Equipment without a current arc flash label (study older than 5 years, or following any modification to upstream protective device settings) shall be treated as PPE Category 4 (maximum) until a new study is completed.

Section 4. Boundaries

Clause 4.2 — The arc flash boundary and limited/restricted approach boundaries shall be physically marked or barricaded before work begins; unqualified personnel are prohibited from crossing the limited approach boundary.

Section 5. Training and Qualification

Clause 5.1 — Only personnel currently qualified per the facility's Electrical Safety Training program may perform energized work or approach within the restricted boundary. Qualification lapses after 36 months without renewal training.

Section 6. Records

Clause 6.1 — Energized work permits and arc flash studies are filed to the Electrical Safety File. Retain permits for 5 years; retain the current arc flash study for the life of the electrical system it covers plus the prior superseded version.